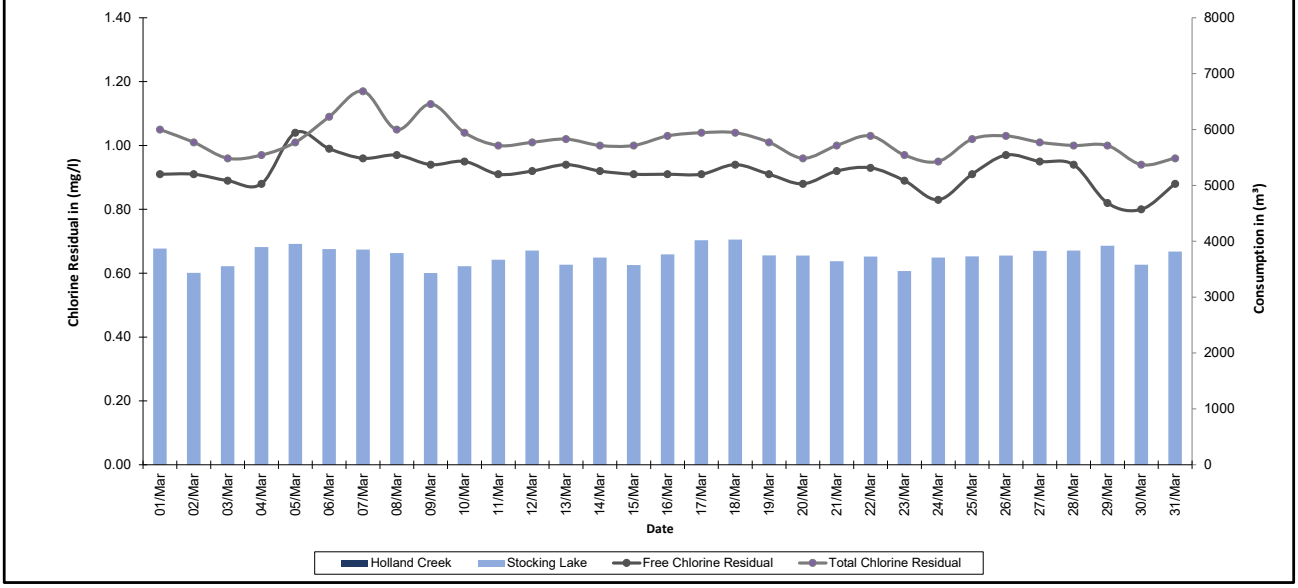
Town of Ladysmith - Arbutus Water treatment Plant

March 2024 - Monthly Report

Date	Daily Flow			Chlorine Residual		CT*	External Lab Testing					
	Stocking Lake	Holland Creek	Combined Flow	Free	Total		HPC	E.coli	Total Coliforms	Aluminum	THM	HAA
	m³	m³	m³	mg/l	mg/l	Minutes·mg/l	CFU	MPN	MPN	mg/l	mg/l	mg/l
01-Mar	3869	0	3869	0.91	1.05	170						
02-Mar	3435	0	3435	0.91	1.01	111						
03-Mar	3555	0	3555	0.89	0.96	134						
04-Mar	3896	0	3896	0.88	0.97	223						
05-Mar	3953	0	3953	1.04	1.01	123	< 1	< 1	< 1	0.0089	0.0162	0.0098
06-Mar	3860	0	3860	0.99	1.09	18						
07-Mar	3851	0	3851	0.96	1.17	123						
08-Mar	3790	0	3790	0.97	1.05	131						
09-Mar	3432	0	3432	0.94	1.13	170						
10-Mar	3555	0	3555	0.95	1.04	132						
11-Mar	3670	0	3670	0.91	1.00	144						
12-Mar	3834	0	3834	0.92	1.01	194						
13-Mar	3580	0	3580	0.94	1.02	198	< 1	< 1	< 1			
14-Mar	3710	0	3710	0.92	1.00	137						
15-Mar	3574	0	3574	0.91	1.00	163						
16-Mar	3764	0	3764	0.91	1.03	158						
17-Mar	4018	0	4018	0.91	1.04	165						
18-Mar	4030	0	4030	0.94	1.04	126						
19-Mar	3746	0	3746	0.91	1.01	122						
20-Mar	3745	0	3745	0.88	0.96	209	0	< 1	< 1			
21-Mar	3642	0	3642	0.92	1.00	70						
22-Mar	3726	0	3726	0.93	1.03	153						
23-Mar	3466	0	3466	0.89	0.97	132						
24-Mar	3708	0	3708	0.83	0.95	183						
25-Mar	3730	0	3730	0.91	1.02	171	< 1	< 1	< 1			
26-Mar	3744	0	3744	0.97	1.03	179						
27-Mar	3829	0	3829	0.95	1.01	144						
28-Mar	3833	0	3833	0.94	1.00	178						
29-Mar	3919	0	3919	0.82	1.00	121						
30-Mar	3581	0	3581	0.80	0.94	184						
31-Mar	3817	0	3817	0.88	0.96	295						
Total	115862	0	115862									
Average	3737	0	3737	0.92	1.02	154	< 0.75	< 1	< 1	0.0089	0.0162	0.00978

*CT - Recorded as the minimum value at the highest daily flow ** Manual Residual were not taken

Arbutus WTP Consumption and Chlorine Residual



VEOLIA

InSight

Town of Ladysmith Arbutus DWTP

Monthly LRV and Turbidity Report

03/01/2024 - 04/01/2024

LRV Monthly Average

Asset	Parameter	Health	Avg	Std. Dev	Points	LL	LCL	%In	% between L and LL	% below LL	Unit
UF 1	LRV		5.0	0.0	32	4.25	4.5	100 %	0 %	0 %	#
UF 2	LRV		5.0	0.0	32	4.25	4.5	100 %	0 %	0 %	#
UF 3	LRV		5.0	0.0	31	4.25	4.5	100 %	0 %	0 %	#

LRV Daily Values

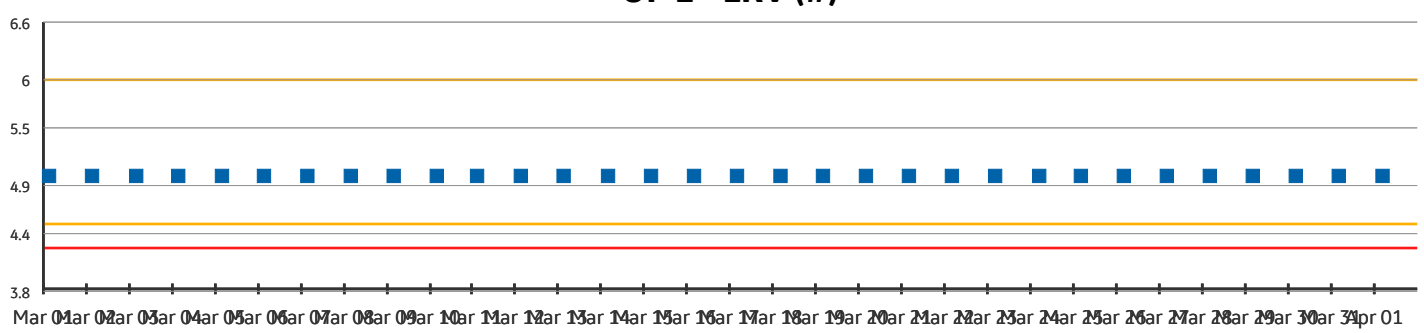
Asset	Parameter	Mar 01	Mar 02	Mar 03	Mar 04	Mar 05	Mar 06	Mar 07	Mar 08	Mar 09	Mar 10
UF 1	LRV	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
UF 2	LRV	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
UF 3	LRV	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0

Asset	Mar 11	Mar 12	Mar 13	Mar 14	Mar 15	Mar 16	Mar 17	Mar 18	Mar 19	Mar 20	Mar 21	Mar 22	Mar 23
UF 1	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
UF 2	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
UF 3	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0

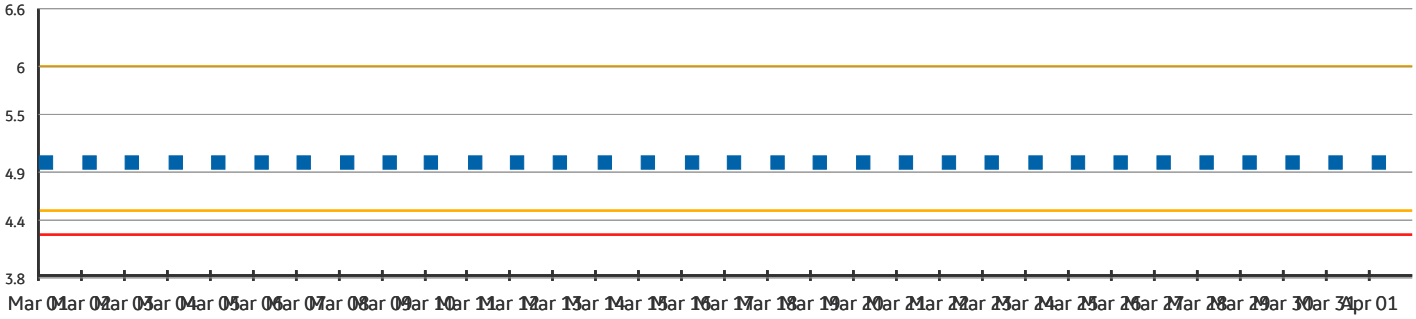
Asset	Mar 24	Mar 25	Mar 26	Mar 27	Mar 28	Mar 29	Mar 30	Mar 31	Apr 01
UF 1	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
UF 2	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
UF 3	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	--

LRV Raw Data

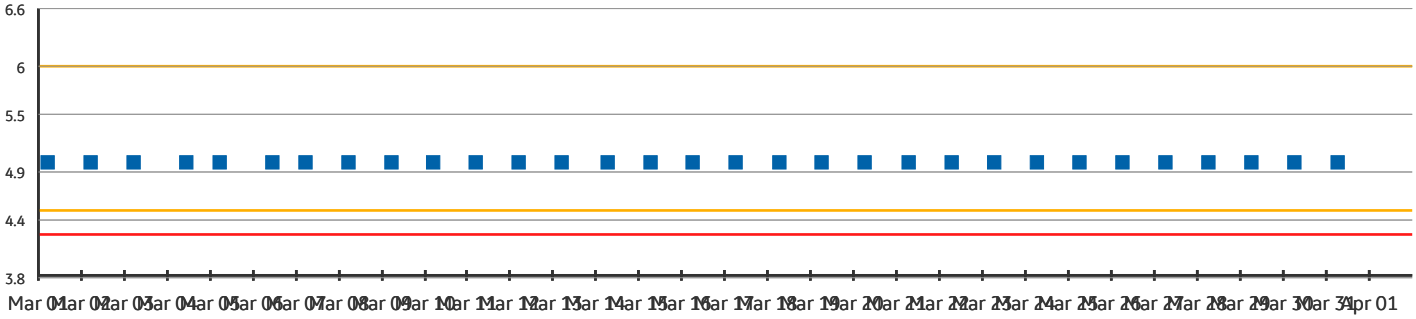
UF 1 - LRV (#)



UF 2 - LRV (#)



UF 3 - LRV (#)



Turbidity Monthly Average

Asset	Parameter	Health	Avg	Std. Dev	Points	UCL	HH	%In	% between H and HH	% above HH	Unit
UF 1	PermeateTurbidity		0.016	0.0	45161	--	--	100 %	0 %	0 %	NTU
UF 2	PermeateTurbidity		0.014	0.0	45161	--	--	100 %	0 %	0 %	NTU
UF 3	PermeateTurbidity		0.015	0.0	45161	--	--	100 %	0 %	0 %	NTU
UF 1	PermeateTurbidityAfterBP		0.016	0.0	429	0.1	0.3	100 %	0 %	0 %	NTU
UF 2	PermeateTurbidityAfterBP		0.014	0.0	416	0.1	0.3	100 %	0 %	0 %	NTU
UF 3	PermeateTurbidityAfterBP		0.015	0.0	419	0.1	0.3	100 %	0 %	0 %	NTU

Turbidity Daily Averages

Asset	Parameter	Mar 01	Mar 02	Mar 03	Mar 04	Mar 05	Mar 06	Mar 07	Mar 08	Mar 09	Mar 10
UF 1	PermeateTurbidity	0.013	0.013	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015
UF 2	PermeateTurbidity	0.013	0.013	0.013	0.013	0.013	0.013	0.013	0.013	0.013	0.013
UF 3	PermeateTurbidity	0.013	0.013	0.013	0.014	0.013	0.014	0.013	0.013	0.013	0.013
UF 1	PermeateTurbidityAfterBP	0.014	0.014	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015
UF 2	PermeateTurbidityAfterBP	0.013	0.013	0.013	0.013	0.013	0.013	0.013	0.013	0.013	0.013
UF 3	PermeateTurbidityAfterBP	0.013	0.013	0.013	0.014	0.013	0.014	0.014	0.014	0.014	0.014

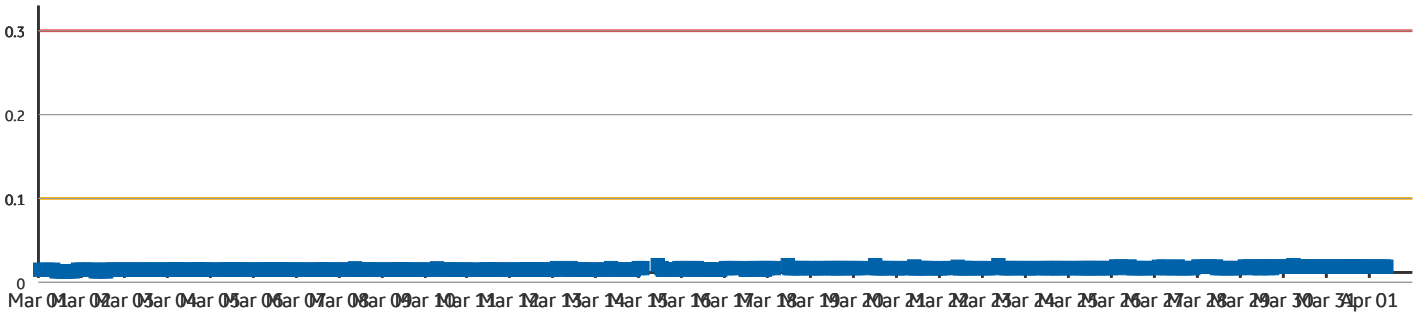
Asset	Mar 11	Mar 12	Mar 13	Mar 14	Mar 15	Mar 16	Mar 17	Mar 18	Mar 19	Mar 20	Mar 21	Mar 22	Mar 23
UF 1	0.015	0.015	0.015	0.015	0.016	0.015	0.016	0.017	0.017	0.017	0.017	0.017	0.017
UF 2	0.013	0.013	0.013	0.013	0.013	0.013	0.013	0.013	0.014	0.015	0.015	0.015	0.015
UF 3	0.013	0.014	0.013	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015
UF 1	0.015	0.015	0.015	0.015	0.016	0.015	0.016	0.017	0.017	0.017	0.017	0.017	0.017
UF 2	0.013	0.014	0.013	0.014	0.013	0.013	0.014	0.014	0.014	0.015	0.015	0.015	0.015

Asset	Mar 11	Mar 12	Mar 13	Mar 14	Mar 15	Mar 16	Mar 17	Mar 18	Mar 19	Mar 20	Mar 21	Mar 22	Mar 23
UF 3	0.014	0.014	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015

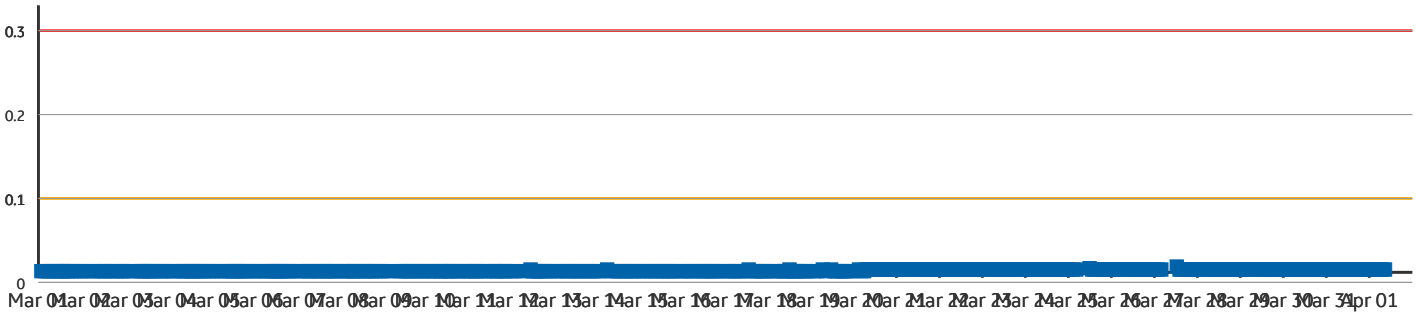
Asset	Mar 24	Mar 25	Mar 26	Mar 27	Mar 28	Mar 29	Mar 30	Mar 31	Apr 01
UF 1	0.017	0.017	0.017	0.017	0.017	0.018	0.019	0.019	0.019
UF 2	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015
UF 3	0.015	0.015	0.015	0.015	0.017	0.017	0.017	0.017	0.017
UF 1	0.017	0.017	0.017	0.017	0.018	0.018	0.019	0.019	0.019
UF 2	0.015	0.015	0.015	0.016	0.015	0.015	0.015	0.015	0.015
UF 3	0.016	0.016	0.016	0.016	0.017	0.017	0.017	0.017	0.017

Turbidity Raw Data

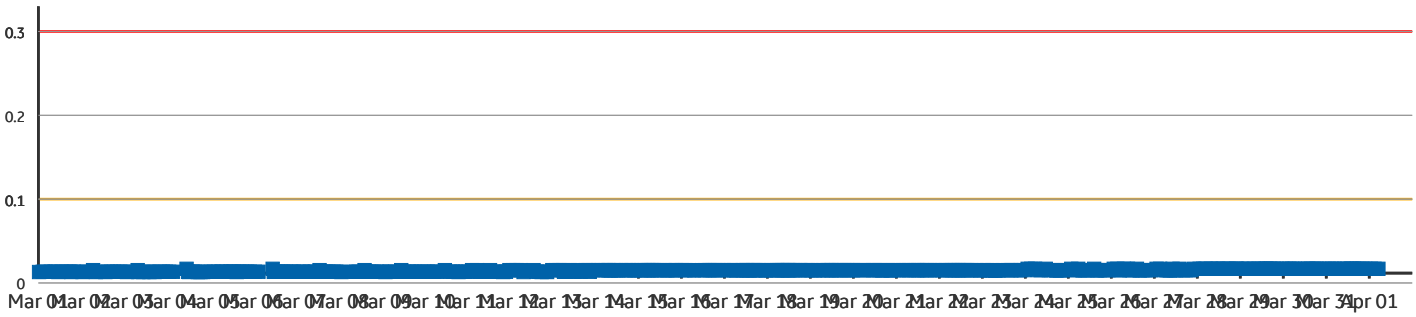
UF 1 - PermeateTurbidityAfterBP (NTU)



UF 2 - PermeateTurbidityAfterBP (NTU)



UF 3 - PermeateTurbidityAfterBP (NTU)



CERTIFICATE OF ANALYSIS

Work Order	: **VA24A4614**	Page	: 1 of 3
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Monthly Sampling	Date Samples Received	: 06-Mar-2024 12:10
PO	: 10880	Date Analysis Commenced	: 07-Mar-2024
C-O-C number	: ----	Issue Date	: 11-Mar-2024 16:52
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 1		
No. of samples analysed	: 1		

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results
- Surrogate Control Limits

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

Signatories	*Position*	*Laboratory Department*
Kevin Duarte	Supervisor - Metals ICP Instrumentation	Metals, Burnaby, British Columbia
Kim Jensen	Department Manager - Metals	Organics, Burnaby, British Columbia
Sanja Risticevic	Department Manager - LCMS	LCMS, Waterloo, Ontario

General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

Where a reported less than (<) result is higher than the LOR, this may be due to primary sample extract/digestate dilution and/or insufficient sample for analysis.

Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.
Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key : CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances
LOR: Limit of Reporting (detection limit).

Unit	*Description*
µg/L	micrograms per litre
mg/L	milligrams per litre

<: less than.

>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.

UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

Analytical Results

Sub-Matrix: **Water**
(Matrix: **Water**)

Analyte	CAS Number	Method/Lab	LOR	Unit	Client sample ID: Treated Water (post reservoir)	----	----	----	----
					Client sampling date / time: 05-Mar-2024 10:30	----	----	----	----
					VA24A4614-001	--------	--------	--------	--------
					Result	----	----	----	----
Total Metals									
Aluminum, total	7429-90-5	E420/VA	0.0030	mg/L	0.0089	----	----	----	----
Volatile Organic Compounds [THMs]									
Bromodichloromethane	75-27-4	E611B/VA	1.0	µg/L	1.2	----	----	----	----
Bromoform	75-25-2	E611B/VA	1.0	µg/L	<1.0	----	----	----	----
Chloroform	67-66-3	E611B/VA	1.0	µg/L	15.0	----	----	----	----
Dibromochloromethane	124-48-1	E611B/VA	1.0	µg/L	<1.0	----	----	----	----
Trihalomethanes [THMs], total	----	E611B/VA	2.0	µg/L	16.2	----	----	----	----
Volatile Organic Compounds [THMs] Surrogates									
Bromofluorobenzene, 4-	460-00-4	E611B/VA	1.0	%	101	----	----	----	----
Difluorobenzene, 1,4-	540-36-3	E611B/VA	1.0	%	97.8	----	----	----	----
Haloacetic Acids									
Bromochloroacetic acid	5589-96-8	E750/WT	1.00	µg/L	<1.00	----	----	----	----
Dibromoacetic acid	631-64-1	E750/WT	1.00	µg/L	<1.00	----	----	----	----
Dichloroacetic acid	79-43-6	E750/WT	1.00	µg/L	4.83	----	----	----	----
Monobromoacetic acid	79-08-3	E750/WT	1.00	µg/L	<1.00	----	----	----	----
Monochloroacetic acid	79-11-8	E750/WT	1.00	µg/L	<1.00	----	----	----	----
Trichloroacetic acid	76-03-9	E750/WT	1.00	µg/L	4.95	----	----	----	----
Haloacetic acids, total [HAA5]	n/a	E750/WT	5.00	µg/L	9.78	----	----	----	----

Please refer to the General Comments section for an explanation of any result qualifiers detected.

Please refer to the Accreditation section for an explanation of analyte accreditations.

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QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: **VA24A4614**	Page	: 1 of 5
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Monthly Sampling	Date Samples Received	: 06-Mar-2024 12:10
PO	: 10880	Issue Date	: 11-Mar-2024 16:51
C-O-C number	: ----		
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 1		
No. of samples analysed	: 1		

This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.

Key

Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.
CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.
DQO: Data Quality Objective.
LOR: Limit of Reporting (detection limit).
RPD: Relative Percent Difference.

Workorder Comments

Holding times are displayed as "---" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

Summary of Outliers

Outliers : Quality Control Samples

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Matrix Spike Duplicate (MSD) outliers occur - please see following pages for full details.
- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- No Analysis Holding Time Outliers exist.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

CERTIFICATE OF ANALYSIS

Work Order	: **VA24A4606**	Page	: 1 of 3
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 06-Mar-2024 12:10
PO	: 10880	Date Analysis Commenced	: 06-Mar-2024
C-O-C number	: ----	Issue Date	: 12-Mar-2024 13:32
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

Signatories	*Position*	*Laboratory Department*
Kate Dimitrova	Supervisor - Inorganic	Inorganics, Burnaby, British Columbia
Kevin Duarte	Supervisor - Metals ICP Instrumentation	Microbiology, Burnaby, British Columbia

General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

Where a reported less than (<) result is higher than the LOR, this may be due to primary sample extract/digestate dilution and/or insufficient sample for analysis.

Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.
Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key : CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances
LOR: Limit of Reporting (detection limit).

Unit	*Description*
µS/cm	microsiemens per centimetre
CFU/mL	colony forming units per millilitre
CU	colour units (1 cu = 1 mg/l pt)
mg/L	milligrams per litre
MPN/100mL	most probable number per hundred millilitres
NTU	nephelometric turbidity units
pH units	pH units

<: less than.

>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.

UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

Workorder Comments

Exceeded Recommended Holding Time prior to receipt at the lab for Microbiology analysis.

Analytical Results

Sub-Matrix: Water (Matrix: Water)				Client sample ID	Raw Water	DAF Effluent	UF Effluent	Treated Water (post reservoir)	----
				Client sampling date / time	05-Mar-2024 10:30	05-Mar-2024 10:30	05-Mar-2024 10:30	05-Mar-2024 10:30	----
Analyte	CAS Number	Method/Lab	LOR	Unit	VA24A4606-001	VA24A4606-002	VA24A4606-003	VA24A4606-004	--------
					Result	Result	Result	Result	----
Physical Tests									
Alkalinity, total (as CaCO3)	----	E290/VA	1.0	mg/L	----	----	----	16.3	----
Colour, true	----	E329/VA	5.0	CU	----	----	----	<5.0	----
Conductivity	----	E100/VA	2.0	µS/cm	----	----	----	67.3	----
pH	----	E108/VA	0.10	pH units	----	----	----	7.55	----
Turbidity	----	E121/VA	0.10	NTU	----	----	----	<0.10	----
Organic / Inorganic Carbon									
Carbon, dissolved organic [DOC]	----	E358-L/VA	0.50	mg/L	2.70	1.13	1.18	----	----
Carbon, total organic [TOC]	----	E355-L/VA	0.50	mg/L	3.92	2.08	1.52	----	----
Microbiological Tests									
Heterotrophic plate count [HPC]	----	E020/VA	1	CFU/mL	----	----	----	<1	----
Coliforms, Escherichia coli [E. coli]	----	E010/VA	1	MPN/100mL	----	----	----	<1	----
Coliforms, total	----	E010/VA	1	MPN/100mL	----	----	----	<1	----

Please refer to the General Comments section for an explanation of any result qualifiers detected.

Please refer to the Accreditation section for an explanation of analyte accreditations.

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QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: **VA24A4606**	Page	: 1 of 7
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 06-Mar-2024 12:10
PO	: 10880	Issue Date	: 12-Mar-2024 13:34
C-O-C number	: ----		
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.

Key

Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.
CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.
DQO: Data Quality Objective.
LOR: Limit of Reporting (detection limit).
RPD: Relative Percent Difference.

Workorder Comments

Holding times are displayed as "---" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

Summary of Outliers

Outliers : Quality Control Samples

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

CERTIFICATE OF ANALYSIS

Work Order	: **VA24A5304**	Page	: 1 of 3
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 14-Mar-2024 10:05
PO	: PO #10880	Date Analysis Commenced	: 14-Mar-2024
C-O-C number	: ----	Issue Date	: 19-Mar-2024 22:48
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

Signatories	*Position*	*Laboratory Department*
Kevin Duarte	Supervisor - Metals ICP Instrumentation	Microbiology, Burnaby, British Columbia
Leon Yang	Analyst	Inorganics, Burnaby, British Columbia
Monica Ko	Lab Assistant	Microbiology, Burnaby, British Columbia

General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

Where a reported less than (<) result is higher than the LOR, this may be due to primary sample extract/digestate dilution and/or insufficient sample for analysis.

Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key : CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances

LOR: Limit of Reporting (detection limit).

Unit	*Description*
µS/cm	microsiemens per centimetre
CFU/mL	colony forming units per millilitre
CU	colour units (1 cu = 1 mg/l pt)
mg/L	milligrams per litre
MPN/100mL	most probable number per hundred millilitres
NTU	nephelometric turbidity units
pH units	pH units

<: less than.

>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.

UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

Analytical Results

Sub-Matrix: **Water**
(Matrix: **Water**)

Analyte	CAS Number	Method/Lab	LOR	Unit	Client sample ID: Raw Water	DAF Effluent	UF Effluent	Treated Water (post reservoir)	----
Client sampling date / time					13-Mar-2024 10:30	13-Mar-2024 10:30	13-Mar-2024 10:30	13-Mar-2024 10:30	----
					VA24A5304-001	VA24A5304-002	VA24A5304-003	VA24A5304-004	--------
					Result	Result	Result	Result	----
Physical Tests									
Alkalinity, total (as CaCO3)	----	E290/VA	1.0	mg/L	----	----	----	15.8	----
Colour, true	----	E329/VA	5.0	CU	----	----	----	<5.0	----
Conductivity	----	E100/VA	2.0	µS/cm	----	----	----	66.2	----
pH	----	E108/VA	0.10	pH units	----	----	----	7.55	----
Turbidity	----	E121/VA	0.10	NTU	----	----	----	<0.10	----
Organic / Inorganic Carbon									
Carbon, dissolved organic [DOC]	----	E358-L/VA	0.50	mg/L	2.74	1.25	1.18	----	----
Carbon, total organic [TOC]	----	E355-L/VA	0.50	mg/L	3.02	1.32	1.10	----	----
Microbiological Tests									
Heterotrophic plate count [HPC]	----	E020/VA	1	CFU/mL	----	----	----	<1	----
Coliforms, Escherichia coli [E. coli]	----	E010/VA	1	MPN/100mL	----	----	----	<1	----
Coliforms, total	----	E010/VA	1	MPN/100mL	----	----	----	<1	----

Please refer to the General Comments section for an explanation of any result qualifiers detected.

Please refer to the Accreditation section for an explanation of analyte accreditations.

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QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: **VA24A5304**	Page	: 1 of 7
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 14-Mar-2024 10:05
PO	: PO #10880	Issue Date	: 19-Mar-2024 22:48
C-O-C number	: ----		
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.

Key

Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.
CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.
DQO: Data Quality Objective.
LOR: Limit of Reporting (detection limit).
RPD: Relative Percent Difference.

Workorder Comments

Holding times are displayed as "---" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

Summary of Outliers

Outliers : Quality Control Samples

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
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- No Matrix Spike outliers occur.
- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

CERTIFICATE OF ANALYSIS

Work Order	: **VA24A5948**	Page	: 1 of 11
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Yearly Sampling	Date Samples Received	: 20-Mar-2024 12:00
PO	: PO #10916	Date Analysis Commenced	: 21-Mar-2024
C-O-C number	: ----	Issue Date	: 09-Apr-2024 16:45
Sampler	: ----		
Site	: ----		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 1		
No. of samples analysed	: 1		

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results
- Surrogate Control Limits

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

Signatories	*Position*	*Laboratory Department*
Amaninder Dhillon	Team Lead - Semi-Volatile Instrumentation	Organics, Waterloo, Ontario
Jon Fisher	Production Manager, Environmental	LCMS, Waterloo, Ontario
Kelly Fischer	Technical Specialist	Inorganics, Waterloo, Ontario
Kevin Duarte	Supervisor - Metals ICP Instrumentation	Metals, Burnaby, British Columbia
Miles Gropen	Department Manager - Inorganics	Inorganics, Burnaby, British Columbia
Natalia Bobretsova		Limnology, Winnipeg, Manitoba
Owen Cheng		Metals, Burnaby, British Columbia
Paul Cushing	Team Leader - Organics	Inorganics, Burnaby, British Columbia
Paul Cushing	Team Leader - Organics	Organics, Burnaby, British Columbia
Sanja Risticevic	Department Manager - LCMS	LCMS, Waterloo, Ontario
Sorina Motea	Laboratory Analyst	Organics, Calgary, Alberta
Stephanie Pinheiro	Team Leader - LCMS	LCMS, Waterloo, Ontario

General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

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Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key : CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances

LOR: Limit of Reporting (detection limit).

Unit	*Description*
µg/L	micrograms per litre
cells/mL	cells per millilitre
mg/L	milligrams per litre
pH units	pH units

<: less than.

>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.

UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

Sample Comments

Sample	Client Id	*Comment*
VA24A5948-001	Treated Water (post resevoir)	No blue-green algae observed.

Qualifiers

Qualifier	*Description*
DLB	*Detection Limit Raised. Analyte detected at comparable level in Method Blank.*
DLM	*Detection Limit Adjusted due to sample matrix effects (e.g. chemical interference, colour, turbidity).*
PEHT	*Parameter exceeded recommended holding time prior to analysis.*

Analytical Results

Sub-Matrix: **Water**
(Matrix: **Water**)

Analyte	CAS Number	Method/Lab	LOR	Unit	Client sample ID: Treated Water (post resevoir)	----	----	----	----
					Client sampling date / time: 19-Mar-2024 10:30	----	----	----	----
					VA24A5948-001	--------	--------	--------	--------
					Result	----	----	----	----
Physical Tests									
Hardness (as CaCO3), from total Ca/Mg	----	EC100A/VA	0.60	mg/L	9.69	----	----	----	----
pH	----	E108/VA	0.10	pH units	7.41	----	----	----	----
Anions and Nutrients									
Bromide	24959-67-9	E235.Br-L/VA	0.050	mg/L	<0.050	----	----	----	----
Chloride	16887-00-6	E235.Cl/VA	0.50	mg/L	8.16	----	----	----	----
Fluoride	16984-48-8	E235.F/VA	0.020	mg/L	<0.020	----	----	----	----
Nitrate (as N)	14797-55-8	E235.NO3-L/VA	0.0050	mg/L	<0.0050	----	----	----	----
Nitrite (as N)	14797-65-0	E235.NO2-L/VA	0.0010	mg/L	<0.0010	----	----	----	----
Sulfate (as SO4)	14808-79-8	E235.SO4/VA	0.30	mg/L	1.97	----	----	----	----
Cyanides									
Cyanide, strong acid dissociable (Total)	----	E333/VA	0.0050	mg/L	<0.0050	----	----	----	----
Taxonomy									
Blue-green algae cell count, total	----	E921A/WP	1	cells/mL	<1	----	----	----	----
Total Metals									
Aluminum, total	7429-90-5	E420/VA	0.0030	mg/L	0.0080	----	----	----	----
Antimony, total	7440-36-0	E420/VA	0.00010	mg/L	<0.00010	----	----	----	----
Arsenic, total	7440-38-2	E420/VA	0.00010	mg/L	<0.00010	----	----	----	----
Barium, total	7440-39-3	E420/VA	0.00010	mg/L	0.00291	----	----	----	----
Beryllium, total	7440-41-7	E420/VA	0.000100	mg/L	<0.000100	----	----	----	----
Bismuth, total	7440-69-9	E420/VA	0.000050	mg/L	<0.000050	----	----	----	----
Boron, total	7440-42-8	E420/VA	0.010	mg/L	0.018	----	----	----	----
Cadmium, total	7440-43-9	E420/VA	0.0000050	mg/L	<0.0000050	----	----	----	----
Calcium, total	7440-70-2	E420/VA	0.050	mg/L	3.14	----	----	----	----
Cesium, total	7440-46-2	E420/VA	0.000010	mg/L	<0.000010	----	----	----	----
Chromium, total	7440-47-3	E420/VA	0.00050	mg/L	<0.00050	----	----	----	----
Cobalt, total	7440-48-4	E420/VA	0.00010	mg/L	<0.00010	----	----	----	----
Copper, total	7440-50-8	E420/VA	0.00050	mg/L	<0.00050	----	----	----	----
Iron, total	7439-89-6	E420/VA	0.010	mg/L	<0.010	----	----	----	----

Analytical Results

Sub-Matrix: Water (Matrix: Water)					Client sample ID	Treated Water (post resevoir)	----	----	----	----
					Client sampling date / time	19-Mar-2024 10:30	----	----	----	----
Analyte	CAS Number	Method/Lab	LOR	Unit		VA24A5948-001	--------	--------	--------	--------
						Result	----	----	----	----
Total Metals										
Lead, total	7439-92-1	E420/VA	0.000050	mg/L		<0.000050	----	----	----	----
Lithium, total	7439-93-2	E420/VA	0.0010	mg/L		<0.0010	----	----	----	----
Magnesium, total	7439-95-4	E420/VA	0.0050	mg/L		0.449	----	----	----	----
Manganese, total	7439-96-5	E420/VA	0.00010	mg/L		0.00056	----	----	----	----
Mercury, total	7439-97-6	E508/VA	0.0000050	mg/L		<0.0000050	----	----	----	----
Molybdenum, total	7439-98-7	E420/VA	0.000050	mg/L		0.000134	----	----	----	----
Nickel, total	7440-02-0	E420/VA	0.00050	mg/L		<0.00050	----	----	----	----
Phosphorus, total	7723-14-0	E420/VA	0.050	mg/L		<0.050	----	----	----	----
Potassium, total	7440-09-7	E420/VA	0.050	mg/L		0.245	----	----	----	----
Rubidium, total	7440-17-7	E420/VA	0.00020	mg/L		0.00057	----	----	----	----
Selenium, total	7782-49-2	E420/VA	0.000050	mg/L		<0.000050	----	----	----	----
Silicon, total	7440-21-3	E420/VA	0.10	mg/L		2.03	----	----	----	----
Silver, total	7440-22-4	E420/VA	0.000010	mg/L		<0.000010	----	----	----	----
Sodium, total	7440-23-5	E420/VA	0.050	mg/L		8.93	----	----	----	----
Strontium, total	7440-24-6	E420/VA	0.00020	mg/L		0.00987	----	----	----	----
Sulfur, total	7704-34-9	E420/VA	0.50	mg/L		0.68	----	----	----	----
Tellurium, total	13494-80-9	E420/VA	0.00020	mg/L		<0.00020	----	----	----	----
Thallium, total	7440-28-0	E420/VA	0.000010	mg/L		<0.000010	----	----	----	----
Thorium, total	7440-29-1	E420/VA	0.00010	mg/L		<0.00010	----	----	----	----
Tin, total	7440-31-5	E420/VA	0.00010	mg/L		<0.00010	----	----	----	----
Titanium, total	7440-32-6	E420/VA	0.00030	mg/L		<0.00030	----	----	----	----
Tungsten, total	7440-33-7	E420/VA	0.00010	mg/L		<0.00010	----	----	----	----
Uranium, total	7440-61-1	E420/VA	0.000010	mg/L		<0.000010	----	----	----	----
Vanadium, total	7440-62-2	E420/VA	0.00050	mg/L		<0.00050	----	----	----	----
Zinc, total	7440-66-6	E420/VA	0.0030	mg/L		<0.0030	----	----	----	----
Zirconium, total	7440-67-7	E420/VA	0.00020	mg/L		<0.00020	----	----	----	----
Volatile Organic Compounds										
Chlorobenzene	108-90-7	E611N/VA	1.0	µg/L		<1.0	----	----	----	----
Chloromethane	74-87-3	E611N/VA	5.0	µg/L		<5.0	----	----	----	----
Dichlorobenzene, 1,2-	95-50-1	E611N/VA	0.50	µg/L		<0.50	----	----	----	----

Analytical Results

Sub-Matrix: Water (Matrix: Water)					Client sample ID	Treated Water (post resevoir)	----	----	----	----
					Client sampling date / time	19-Mar-2024 10:30	----	----	----	----
Analyte	CAS Number	Method/Lab	LOR	Unit	VA24A5948-001	--------	--------	--------	--------	
					Result	----	----	----	----	
Volatile Organic Compounds										
Dichlorobenzene, 1,3-	541-73-1	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Dichlorobenzene, 1,4-	106-46-7	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Dichloropropane, 1,2-	78-87-5	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Dichloropropylene, cis+trans-1,3-	542-75-6	E611N/VA	1.5	µg/L	<1.5	----	----	----	----	
Dichloropropylene, cis-1,3-	10061-01-5	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Tetrachloroethane, 1,1,1,2-	630-20-6	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Tetrachloroethane, 1,1,2,2-	79-34-5	E611N/VA	0.20	µg/L	<0.20	----	----	----	----	
Trichloroethane, 1,1,2-	79-00-5	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Trichlorofluoromethane	75-69-4	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Volatile Organic Compounds [Drycleaning]										
Carbon tetrachloride	56-23-5	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Chloroethane	75-00-3	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Dichloroethane, 1,1-	75-34-3	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Dichloroethylene, 1,1-	75-35-4	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Dichloroethylene, cis-1,2-	156-59-2	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Dichloroethylene, trans-1,2-	156-60-5	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Dichloromethane	75-09-2	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Dichloropropylene, trans-1,3-	10061-02-6	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Tetrachloroethylene	127-18-4	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Trichloroethane, 1,1,1-	71-55-6	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Trichloroethylene	79-01-6	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Vinyl chloride	75-01-4	E611N/VA	0.40	µg/L	<0.40	----	----	----	----	
Volatile Organic Compounds [Fuels]										
Benzene	71-43-2	E611N/VA	0.50	µg/L	<0.50	----	----	----	----	
Butadiene, 1,3-	106-99-0	E611N/VA	0.20	µg/L	<0.20	----	----	----	----	
Cymene, p-	99-87-6	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Decane, n-	124-18-5	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Dibromoethane, 1,2-	106-93-4	E611N/VA	0.20	µg/L	<0.20	----	----	----	----	
Dichloroethane, 1,2-	107-06-2	E611N/VA	1.0	µg/L	<1.0	----	----	----	----	
Ethylbenzene	100-41-4	E611N/VA	0.50	µg/L	<0.50	----	----	----	----	

Analytical Results

Sub-Matrix: Water (Matrix: Water)					Client sample ID	Treated Water (post resevoir)	----	----	----	----
					Client sampling date / time	19-Mar-2024 10:30	----	----	----	----
Analyte	CAS Number	Method/Lab	LOR	Unit		VA24A5948-001	--------	--------	--------	--------
						Result	----	----	----	----
Volatile Organic Compounds [Fuels]										
Hexane, n-	110-54-3	E611N/VA	1.0	µg/L		<1.0	----	----	----	----
Isopropylbenzene	98-82-8	E611N/VA	1.0	µg/L		<1.0	----	----	----	----
Methylcyclohexane	108-87-2	E611N/VA	1.0	µg/L		<1.0	----	----	----	----
Methyl-tert-butyl ether [MTBE]	1634-04-4	E611N/VA	0.50	µg/L		<0.50	----	----	----	----
Propylbenzene, n-	103-65-1	E611N/VA	1.0	µg/L		<1.0	----	----	----	----
Styrene	100-42-5	E611N/VA	0.50	µg/L		<0.50	----	----	----	----
Toluene	108-88-3	E611N/VA	0.40	µg/L		<0.40	----	----	----	----
Trimethylbenzene, 1,2,4-	95-63-6	E611N/VA	1.0	µg/L		<1.0	----	----	----	----
Trimethylbenzene, 1,3,5-	108-67-8	E611N/VA	1.0	µg/L		<1.0	----	----	----	----
Xylene, m+p-	179601-23-1	E611N/VA	0.40	µg/L		<0.40	----	----	----	----
Xylene, o-	95-47-6	E611N/VA	0.30	µg/L		<0.30	----	----	----	----
Xylenes, total	1330-20-7	E611N/VA	0.50	µg/L		<0.50	----	----	----	----
Volatile Organic Compounds [THMs]										
Bromodichloromethane	75-27-4	E611N/VA	1.0	µg/L		1.4	----	----	----	----
Bromoform	75-25-2	E611N/VA	1.0	µg/L		<1.0	----	----	----	----
Chloroform	67-66-3	E611N/VA	1.0	µg/L		11.7	----	----	----	----
Dibromochloromethane	124-48-1	E611N/VA	1.0	µg/L		<1.0	----	----	----	----
Hydrocarbons										
VHw (C6-C10)	----	E581.VH+F1/VA	100	µg/L		<100	----	----	----	----
VPHw	----	EC580A/VA	100	µg/L		<100	----	----	----	----
Hydrocarbons Surrogates										
Dichlorotoluene, 3,4-	95-75-0	E581.VH+F1/VA	1.0	%		105	----	----	----	----
Volatile Organic Compounds Surrogates										
Bromofluorobenzene, 4-	460-00-4	E611N/VA	1.0	%		98.6	----	----	----	----
Difluorobenzene, 1,4-	540-36-3	E611N/VA	1.0	%		102	----	----	----	----
Polycyclic Aromatic Hydrocarbons										
Acenaphthene	83-32-9	E641A/CG	0.010	µg/L		<0.010	----	----	----	----
Acenaphthylene	208-96-8	E641A/CG	0.010	µg/L		<0.010	----	----	----	----

Analytical Results

Sub-Matrix: Water (Matrix: Water)					Client sample ID	Treated Water (post resevoir)	----	----	----	----
					Client sampling date / time	19-Mar-2024 10:30	----	----	----	----
Analyte	CAS Number	Method/Lab	LOR	Unit	VA24A5948-001	--------	--------	--------	--------	
					Result	----	----	----	----	
Polycyclic Aromatic Hydrocarbons										
Acridine	260-94-6	E641A/CG	0.010	µg/L	<0.010	----	----	----	----	
Anthracene	120-12-7	E641A/CG	0.010	µg/L	<0.010	----	----	----	----	
Benz(a)anthracene	56-55-3	E641A/CG	0.010	µg/L	<0.010	----	----	----	----	
Benzo(a)pyrene	50-32-8	E641A/CG	0.0050	µg/L	<0.0050	----	----	----	----	
Benzo(b+j)fluoranthene	n/a	E641A/CG	0.010	µg/L	<0.010	----	----	----	----	
Benzo(b+j+k)fluoranthene	n/a	E641A/CG	0.015	µg/L	<0.015	----	----	----	----	
Benzo(g,h,i)perylene	191-24-2	E641A/CG	0.010	µg/L	<0.010	----	----	----	----	
Benzo(k)fluoranthene	207-08-9	E641A/CG	0.010	µg/L	<0.010	----	----	----	----	
Chrysene	218-01-9	E641A/CG	0.010	µg/L	<0.010	----	----	----	----	
Dibenz(a,h)anthracene	53-70-3	E641A/CG	0.0050	µg/L	<0.0050	----	----	----	----	
Fluoranthene	206-44-0	E641A/CG	0.010	µg/L	<0.010	----	----	----	----	
Fluorene	86-73-7	E641A/CG	0.010	µg/L	<0.010	----	----	----	----	
Indeno(1,2,3-c,d)pyrene	193-39-5	E641A/CG	0.010	µg/L	<0.010	----	----	----	----	
Methylnaphthalene, 1-	90-12-0	E641A/CG	0.010	µg/L	<0.010	----	----	----	----	
Methylnaphthalene, 1+2-	----	E641A/CG	0.015	µg/L	<0.015	----	----	----	----	
Methylnaphthalene, 2-	91-57-6	E641A/CG	0.010	µg/L	<0.010	----	----	----	----	
Naphthalene	91-20-3	E641A/CG	0.050	µg/L	<0.050	----	----	----	----	
Phenanthrene	85-01-8	E641A/CG	0.020	µg/L	<0.020	----	----	----	----	
Pyrene	129-00-0	E641A/CG	0.010	µg/L	<0.010	----	----	----	----	
Quinoline	91-22-5	E641A/CG	0.050	µg/L	<0.050	----	----	----	----	
B(a)P total potency equivalents [B(a)P TPE]	----	E641A/CG	0.010	µg/L	<0.010	----	----	----	----	
PAHs, high molecular weight (BC AWQ)	n/a	E641A/CG	0.030	µg/L	<0.030	----	----	----	----	
PAHs, low molecular weight (BC AWQ)	n/a	E641A/CG	0.060	µg/L	<0.060	----	----	----	----	
PAHs, total (EPA 16)	n/a	E641A/CG	0.065	µg/L	<0.065	----	----	----	----	
Polycyclic Aromatic Hydrocarbons Surrogates										
Chrysene-d12	1719-03-5	E641A/CG	0.1	%	106	----	----	----	----	
Naphthalene-d8	1146-65-2	E641A/CG	0.1	%	123	----	----	----	----	
Phenanthrene-d10	1517-22-2	E641A/CG	0.1	%	101	----	----	----	----	
Disinfectant By-Products										
Bromate	15541-45-4	E722A/WT	0.30	µg/L	0.39	----	----	----	----	

Analytical Results

Sub-Matrix: Water (Matrix: Water)					Client sample ID	Treated Water (post resevoir)	----	----	----	----
					Client sampling date / time	19-Mar-2024 10:30	----	----	----	----
Analyte	CAS Number	Method/Lab	LOR	Unit		VA24A5948-001	--------	--------	--------	--------
						Result	----	----	----	----
DIsinfectant By-Products										
Chlorate	14866-68-3	E409.CLO3/WT	0.010	mg/L		0.022	----	----	----	----
Chlorite	14998-27-7	E409.CLO2/WT	0.010	mg/L		<0.010	----	----	----	----
Semi-Volatile Organics										
Nitrosodimethylamine, N- [NDMA]	62-75-9	E725-T/WT	0.00090	µg/L		<0.00135 DLB	----	----	----	----
Per- and Perfluoroalkyl Substances (PFAS)										
Perfluorooctanesulfonic acid [PFOS]	1763-23-1	E745/WT	0.020	µg/L		<0.020	----	----	----	----
Perfluorooctanoic acid [PFOA]	335-67-1	E745/WT	0.020	µg/L		<0.020	----	----	----	----
Per- and Perfluoroalkyl Substances (PFAS) Surrogates										
Perfluorooctanesulfonic acid [13C8-PFOS]	265893-05-6	E745/WT	0.01	%		96.0	----	----	----	----
Chlorinated Phenolics										
Chlorophenol, 2-	95-57-8	E651A/VA	0.050	µg/L		<0.050	----	----	----	----
Chlorophenol, 3-	108-43-0	E651A/VA	0.050	µg/L		<0.050	----	----	----	----
Chlorophenol, 4-	106-48-9	E651A/VA	0.050	µg/L		<0.050	----	----	----	----
Dichlorophenol, 2,3-	576-24-9	E651A/VA	0.050	µg/L		<0.050	----	----	----	----
Dichlorophenol, 2,4- + 2,5-	----	E651A/VA	0.050	µg/L		<0.050	----	----	----	----
Dichlorophenol, 2,6-	87-65-0	E651A/VA	0.050	µg/L		<0.050	----	----	----	----
Dichlorophenol, 3,4-	95-77-2	E651A/VA	0.050	µg/L		<0.050	----	----	----	----
Dichlorophenol, 3,5-	591-35-5	E651A/VA	0.050	µg/L		<0.050	----	----	----	----
Methylphenol, 4-chloro-3-	59-50-7	E651A/VA	0.10	µg/L		<0.10	----	----	----	----
Pentachlorophenol [PCP]	87-86-5	E651A/VA	0.10	µg/L		<0.10	----	----	----	----
Tetrachlorophenol, 2,3,4,5-	4901-51-3	E651A/VA	0.10	µg/L		<0.10	----	----	----	----
Tetrachlorophenol, 2,3,4,6-	58-90-2	E651A/VA	0.10	µg/L		<0.10	----	----	----	----
Tetrachlorophenol, 2,3,5,6-	935-95-5	E651A/VA	0.10	µg/L		<0.10	----	----	----	----
Trichlorophenol, 2,3,4-	15950-66-0	E651A/VA	0.10	µg/L		<0.10	----	----	----	----
Trichlorophenol, 2,3,5-	933-78-8	E651A/VA	0.10	µg/L		<0.10	----	----	----	----
Trichlorophenol, 2,3,6-	933-75-5	E651A/VA	0.10	µg/L		<0.10	----	----	----	----
Trichlorophenol, 2,4,5-	95-95-4	E651A/VA	0.10	µg/L		<0.10	----	----	----	----
Trichlorophenol, 2,4,6-	88-06-2	E651A/VA	0.10	µg/L		<0.10	----	----	----	----

Analytical Results

Sub-Matrix: Water (Matrix: Water)					Client sample ID	Treated Water (post resevoir)	----	----	----	----
					Client sampling date / time	19-Mar-2024 10:30	----	----	----	----
Analyte	CAS Number	Method/Lab	LOR	Unit		VA24A5948-001	--------	--------	--------	--------
						Result	----	----	----	----
Chlorinated Phenolics										
Trichlorophenol, 3,4,5-	609-19-8	E651A/VA	0.10	µg/L		<0.10	----	----	----	----
Phenolics Surrogates										
Chlorophenol-d4, 2-	93951-73-6	E651A/VA	0.1	%		96.4	----	----	----	----
Dichlorophenol-d3, 2,4-	93951-74-7	E651A/VA	0.1	%		91.8	----	----	----	----
Tribromophenol, 2,4,6-	118-79-6	E651A/VA	0.1	%		97.4	----	----	----	----
Organochlorine Pesticides										
Toxaphene, total	8001-35-2	E660C/WT	0.50	µg/L		<0.50	----	----	----	----
Organochlorine Pesticides Surrogates										
Decachlorobiphenyl	2051-24-3	E660C/WT	0.1	%		89.0	----	----	----	----
Tetrachloro-m-xylene	877-09-8	E660C/WT	0.1	%		72.8	----	----	----	----
Herbicides										
Acetic acid, 2-methyl-4-chlorophenoxy- [MCPA]	94-74-6	E706A/WT	0.050	µg/L		<0.050	----	----	----	----
Alachlor	15972-60-8	E755/WT	0.050	µg/L		<0.050	----	----	----	----
Ametryn	834-12-8	E755/WT	0.0250	µg/L		<0.0250	----	----	----	----
Asulam	3337-71-1	E706A/WT	0.010	µg/L		<0.010	----	----	----	----
Atrazine	1912-24-9	E755/WT	0.050	µg/L		<0.050	----	----	----	----
Atrazine + N-dealkylated metabolites	----	E755/WT	0.10	µg/L		<0.10	----	----	----	----
Atrazine-desethyl	6190-65-4	E755/WT	0.0250	µg/L		<0.0250	----	----	----	----
Brodifacoum	56073-10-0	E706A/WT	0.010	µg/L		<0.100 DLM	----	----	----	----
Bromacil	314-40-9	E706A/WT	0.10	µg/L		<0.10	----	----	----	----
Bromoxynil	1689-84-5	E706A/WT	0.050	µg/L		<0.050	----	----	----	----
Butanoic acid, 4-(4-chloro-2-methylphenoxy)- [MCPB]	94-81-5	E706A/WT	0.050	µg/L		<0.050	----	----	----	----
Clopyralid	1702-17-6	E706A/WT	0.10	µg/L		<0.10	----	----	----	----
Cyanazine	21725-46-2	E755/WT	0.100	µg/L		<0.100	----	----	----	----
Dicamba	1918-00-9	E706A/WT	0.10	µg/L		<0.10	----	----	----	----
Dichlorophenoxy(2,4-)butyric acid, 4- [2,4-DB]	94-82-6	E706A/WT	0.050	µg/L		<0.050	----	----	----	----
Dichlorophenoxyacetic acid, 2,4- [2,4-D]	94-75-7	E706A/WT	0.050	µg/L		<0.050	----	----	----	----
Dichlorprop [2,4-DP]	120-36-5	E706A/WT	0.050	µg/L		<0.050	----	----	----	----
Diclofop-methyl	51338-27-3	E755/WT	0.100	µg/L		<0.100	----	----	----	----

Analytical Results

Sub-Matrix: Water (Matrix: Water)					Client sample ID	Treated Water (post resevoir)	----	----	----	----
					Client sampling date / time	19-Mar-2024 10:30	----	----	----	----
Analyte	CAS Number	Method/Lab	LOR	Unit		VA24A5948-001	--------	--------	--------	--------
						Result	----	----	----	----
Herbicides										
Diflufenican	83164-33-4	E706A/WT	0.010	µg/L		<0.010	----	----	----	----
Dinoseb	88-85-7	E706A/WT	0.050	µg/L		<0.050	----	----	----	----
Diquat (ion)	2764-72-9	E723A/WT	1.0	µg/L		<1.0	----	----	----	----
Ethalfluralin	55283-68-6	E756/WT	0.10	µg/L		<0.10 PEHT	----	----	----	----
Fluazifop-p-butyl	79241-46-6	E755/WT	0.0250	µg/L		<0.0250	----	----	----	----
Glyphosate	1071-83-6	E716B-L/WT	0.10	µg/L		<0.10	----	----	----	----
Linuron	330-55-2	E706A/WT	0.10	µg/L		<0.10	----	----	----	----
Mecoprop [MCPP]	95-65-2	E706A/WT	0.050	µg/L		<0.050	----	----	----	----
Metolachlor	51218-45-2	E755/WT	0.0250	µg/L		<0.0250	----	----	----	----
Metribuzin	21087-64-9	E755/WT	0.100	µg/L		<0.100	----	----	----	----
Nicarbazin	330-95-0	E706A/WT	0.010	µg/L		<0.010	----	----	----	----
Oryzalin	19044-88-3	E706A/WT	0.050	µg/L		<0.050	----	----	----	----
Paraquat (as dichloride)	1910-42-5	E723A/WT	1.0	µg/L		<1.0	----	----	----	----
Picloram	1918-02-1	E706A/WT	0.10	µg/L		<0.10	----	----	----	----
Prometon	1610-18-0	E755/WT	0.0250	µg/L		<0.0250	----	----	----	----
Prometryn	7287-19-6	E755/WT	0.0250	µg/L		<0.0250	----	----	----	----
Propanil	709-98-8	E706A/WT	0.010	µg/L		<0.010	----	----	----	----
Propazine	139-40-2	E755/WT	0.0250	µg/L		<0.0250	----	----	----	----
Simazine	122-34-9	E755/WT	0.100	µg/L		<0.100	----	----	----	----
Terbacil	5902-51-2	E706A/WT	0.010	µg/L		<0.010	----	----	----	----
Terbutryn	886-50-0	E755/WT	0.0250	µg/L		<0.0250	----	----	----	----
Triallate	2303-17-5	E755/WT	0.100	µg/L		<0.100	----	----	----	----
Trichlorophenoxyacetic acid, 2,4,5- [2,4,5-T]	93-76-5	E706A/WT	0.050	µg/L		<0.050	----	----	----	----
Trichlorophenoxypropionic acid, 2,4,5- [2,4,5-TP]	93-72-1	E706A/WT	0.050	µg/L		<0.050	----	----	----	----
Triclopyr	55335-06-3	E706A/WT	0.050	µg/L		<0.050	----	----	----	----
Trifluralin	1582-09-8	E756/WT	0.10	µg/L		<0.10 PEHT	----	----	----	----
Herbicides Surrogates										
Dichlorophenylacetic acid, 2,4-	19719-28-9	E706A/WT	1.0	%		108	----	----	----	----
Insecticides										

Analytical Results

Sub-Matrix: Water (Matrix: Water)					Client sample ID	Treated Water (post resevoir)	----	----	----	----
					Client sampling date / time	19-Mar-2024 10:30	----	----	----	----
Analyte	CAS Number	Method/Lab	LOR	Unit		VA24A5948-001	--------	--------	--------	--------
						Result	----	----	----	----
Insecticides										
Azinphos-methyl	86-50-0	E755/WT	0.100	µg/L		<0.100	----	----	----	----
Bendiocarb	22781-23-3	E755/WT	0.0250	µg/L		<0.0250	----	----	----	----
Carbaryl	63-25-2	E755/WT	0.050	µg/L		<0.050	----	----	----	----
Carbofuran	1563-66-2	E755/WT	0.0250	µg/L		<0.0250	----	----	----	----
Chlorpyrifos	2921-88-2	E756/WT	0.10	µg/L		<0.10 PEHT	----	----	----	----
Diazinon	333-41-5	E755/WT	0.0250	µg/L		<0.0250	----	----	----	----
Dimethoate	60-51-5	E755/WT	0.050	µg/L		<0.050	----	----	----	----
Malathion	121-75-5	E755/WT	0.0250	µg/L		<0.0250	----	----	----	----
Parathion	56-38-2	E756/WT	0.10	µg/L		<0.10 PEHT	----	----	----	----
Parathion-methyl	298-00-0	E756/WT	0.10	µg/L		<0.10 PEHT	----	----	----	----
Phorate	298-02-2	E755/WT	0.250	µg/L		<0.250	----	----	----	----
Temephos	3383-96-8	E755/WT	0.250	µg/L		<0.250	----	----	----	----
Terbufos	13071-79-9	E755/WT	0.50	µg/L		<0.50	----	----	----	----
Nitrosamines Surrogates										
Nitrosodimethylamine-d6, N-	17829-05-9	E725-T/WT	0.00090	%		97.2	----	----	----	----

Please refer to the General Comments section for an explanation of any result qualifiers detected.

Please refer to the Accreditation section for an explanation of analyte accreditations.

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QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: **VA24A5948**	Page	: 1 of 15
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Yearly Sampling	Date Samples Received	: 20-Mar-2024 12:00
PO	: PO #10916	Issue Date	: 09-Apr-2024 16:45
C-O-C number	: ----		
Sampler	: ----		
Site	: ----		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 1		
No. of samples analysed	: 1		

This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.

Key

Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.
CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.
DQO: Data Quality Objective.
LOR: Limit of Reporting (detection limit).
RPD: Relative Percent Difference.

Workorder Comments

Holding times are displayed as "---" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

Summary of Outliers

Outliers : Quality Control Samples

- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- Method Blank value outliers occur - please see following pages for full details.
- Matrix Spike outliers occur - please see following pages for full details.
- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

CERTIFICATE OF ANALYSIS

Work Order	: **VA24A5953**	Page	: 1 of 3
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 21-Mar-2024 10:00
PO	: PO #10916	Date Analysis Commenced	: 21-Mar-2024
C-O-C number	: ----	Issue Date	: 27-Mar-2024 17:24
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

Signatories	*Position*	*Laboratory Department*
Kevin Duarte	Supervisor - Metals ICP Instrumentation	Microbiology, Burnaby, British Columbia
Kim Jensen	Department Manager - Metals	Inorganics, Burnaby, British Columbia

General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

Where a reported less than (<) result is higher than the LOR, this may be due to primary sample extract/digestate dilution and/or insufficient sample for analysis.

Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key : CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances

LOR: Limit of Reporting (detection limit).

Unit	*Description*
µS/cm	microsiemens per centimetre
CFU/mL	colony forming units per millilitre
CU	colour units (1 cu = 1 mg/l pt)
mg/L	milligrams per litre
MPN/100mL	most probable number per hundred millilitres
NTU	nephelometric turbidity units
pH units	pH units

<: less than.

>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.

UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

Qualifiers

Qualifier	*Description*
SFP	*Sample was filtered and preserved at the laboratory.*

Analytical Results

Sub-Matrix: **Water**
(Matrix: **Water**)

Analyte	CAS Number	Method/Lab	LOR	Unit	Client sample ID: Raw Water	DAF Effluent	UF Effluent	Treated Water (post reservoir)	----
Client sampling date / time					20-Mar-2024 10:30	20-Mar-2024 10:30	20-Mar-2024 10:30	20-Mar-2024 10:30	----
					VA24A5953-001	VA24A5953-002	VA24A5953-003	VA24A5953-004	--------
					Result	Result	Result	Result	----
Physical Tests									
Alkalinity, total (as CaCO3)	----	E290/VA	1.0	mg/L	----	----	----	16.0	----
Colour, true	----	E329/VA	5.0	CU	----	----	----	<5.0	----
Conductivity	----	E100/VA	2.0	µS/cm	----	----	----	65.1	----
pH	----	E108/VA	0.10	pH units	----	----	----	7.30	----
Turbidity	----	E121/VA	0.10	NTU	----	----	----	<0.10	----
Organic / Inorganic Carbon									
Carbon, dissolved organic [DOC]	----	E358-L/VA	0.50	mg/L	3.31	1.23 SFP	1.25 SFP	----	----
Carbon, total organic [TOC]	----	E355-L/VA	0.50	mg/L	2.94	1.34	1.30	----	----
Microbiological Tests									
Heterotrophic plate count [HPC]	----	E020/VA	1	CFU/mL	----	----	----	0	----
Coliforms, Escherichia coli [E. coli]	----	E010/VA	1	MPN/100mL	----	----	----	<1	----
Coliforms, total	----	E010/VA	1	MPN/100mL	----	----	----	<1	----

Please refer to the General Comments section for an explanation of any result qualifiers detected.

Please refer to the Accreditation section for an explanation of analyte accreditations.

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QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: **VA24A5953**	Page	: 1 of 8
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 21-Mar-2024 10:00
PO	: PO #10916	Issue Date	: 27-Mar-2024 17:20
C-O-C number	: ----		
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.

Key

Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.
CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.
DQO: Data Quality Objective.
LOR: Limit of Reporting (detection limit).
RPD: Relative Percent Difference.

Workorder Comments

Holding times are displayed as "---" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

Summary of Outliers

Outliers : Quality Control Samples

- No Method Blank value outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- Duplicate outliers occur - please see following pages for full details.
- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- Quality Control Sample Frequency Outliers occur - please see following pages for full details.

CERTIFICATE OF ANALYSIS

Work Order	: **VA24A6302**	Page	: 1 of 3
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 26-Mar-2024 12:00
PO	: 10880	Date Analysis Commenced	: 26-Mar-2024
C-O-C number	: ----	Issue Date	: 02-Apr-2024 08:44
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

Signatories	*Position*	*Laboratory Department*
Brieanna Allen	Production/Validation Manager	Microbiology, Burnaby, British Columbia
Kim Jensen	Department Manager - Metals	Inorganics, Burnaby, British Columbia
Miles Gropen	Department Manager - Inorganics	Inorganics, Burnaby, British Columbia
Miles Gropen	Department Manager - Inorganics	Microbiology, Burnaby, British Columbia

General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

Where a reported less than (<) result is higher than the LOR, this may be due to primary sample extract/digestate dilution and/or insufficient sample for analysis.

Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key : CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances

LOR: Limit of Reporting (detection limit).

Unit	*Description*
µS/cm	microsiemens per centimetre
CFU/mL	colony forming units per millilitre
CU	colour units (1 cu = 1 mg/l pt)
mg/L	milligrams per litre
MPN/100mL	most probable number per hundred millilitres
NTU	nephelometric turbidity units
pH units	pH units

<: less than.

>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.

UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

Workorder Comments

Sample(s) 004: Exceeded Recommended Holding Time prior to receipt at the lab for HPC analysis.

Qualifiers

Qualifier	*Description*
HTD	*Hold time exceeded for re-analysis or dilution, but initial testing was conducted within hold time.*
SFP	*Sample was filtered and preserved at the laboratory.*

Analytical Results

Sub-Matrix: **Water**
(Matrix: **Water**)

Analyte	CAS Number	Method/Lab	LOR	Unit	Client sample ID: Raw Water	DAF Effluent	UF Effluent	Treated Water (post reservoir)	----
Client sampling date / time					25-Mar-2024 08:30	25-Mar-2024 08:30	25-Mar-2024 08:30	25-Mar-2024 08:30	----
					VA24A6302-001	VA24A6302-002	VA24A6302-003	VA24A6302-004	--------
					Result	Result	Result	Result	----
Physical Tests									
Alkalinity, total (as CaCO3)	----	E290/VA	1.0	mg/L	----	----	----	15.8	----
Colour, true	----	E329/VA	5.0	CU	----	----	----	<5.0	----
Conductivity	----	E100/VA	2.0	µS/cm	----	----	----	65.6	----
pH	----	E108/VA	0.10	pH units	----	----	----	7.40	----
Turbidity	----	E121/VA	0.10	NTU	----	----	----	<0.10	----
Organic / Inorganic Carbon									
Carbon, dissolved organic [DOC]	----	E358-L/VA	0.50	mg/L	2.76 HTD, SFP	1.35 HTD, SFP	1.25 HTD, SFP	----	----
Carbon, total organic [TOC]	----	E355-L/VA	0.50	mg/L	2.40	1.21	1.00	----	----
Microbiological Tests									
Heterotrophic plate count [HPC]	----	E020/VA	1	CFU/mL	----	----	----	<1	----
Coliforms, Escherichia coli [E. coli]	----	E010/VA	1	MPN/100mL	----	----	----	<1	----
Coliforms, total	----	E010/VA	1	MPN/100mL	----	----	----	<1	----

Please refer to the General Comments section for an explanation of any result qualifiers detected.

Please refer to the Accreditation section for an explanation of analyte accreditations.

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QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: **VA24A6302**	Page	: 1 of 7
Client	: **Town of Ladysmith**	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
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PO	: 10880	Issue Date	: 02-Apr-2024 08:44
C-O-C number	: ----		
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

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Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.